MODULE V :Open Stack Compute
Open Stack Compute -The compute service components - Deciding on the hypervisor – Open Stack Magnum Project - Segregating the compute cloud – Over commitment considerations - Storing instances' alternatives - Understanding instance booting - Planning for service recovery.

OpenStack Compute Service Components

The core components of the OpenStack compute service include:

- **Nova (Compute Service):** The conductor for the VM lifecycle, handling tasks like provisioning, starting, stopping, and managing VMs.
- **Cinder (Block Storage Service):** Provides persistent block storage volumes that function like virtual hard disks for VMs.
- **Glance (Image Service):** The central repository for storing virtual machine images used to create VMs.

Deciding on the Hypervisor

A hypervisor is software that allows you to run multiple virtual machines on a single physical server. OpenStack supports various hypervisors, including:

- **KVM (Kernel-based Virtual Machine):** A widely used open-source hypervisor known for its efficiency and security.
- **VMware ESXi:** A popular enterprise-grade hypervisor offering advanced features but requiring a commercial license.
- **Hyper-V:** Microsoft's hypervisor solution known for tight integration with Windows environments.

Choosing the right hypervisor depends on factors like:

- **Performance requirements:** Consider the resource demands of your workloads and the hypervisor's capabilities.
- **Cost:** Evaluate open-source vs. commercial licensing costs.
- **Skillset:** Choose a hypervisor your team is familiar with for easier management.
- **Features:** Identify any specific features required by your applications.

OpenStack Magnum Project

Magnum is an optional OpenStack project that provides functionalities for provisioning and managing container orchestration engines like Docker Swarm or Kubernetes. It allows you to integrate containerization technologies with your OpenStack cloud environment.

Segregating the Compute Cloud

There are several ways to segregate your compute cloud, including:

- **Tenancy:** OpenStack uses projects to segregate resources for different users or departments.
- **Security Groups:** These act like virtual firewalls, allowing you to control network traffic flow and isolate workloads.
- **Availability Zones:** You can distribute VMs across physically separate zones for fault tolerance and redundancy.

Overcommitment Considerations

Overcommitment refers to allocating more resources (CPU, memory) than you physically have, relying on statistical multiplexing to avoid resource exhaustion.

Benefits:

- Improved resource utilization

Challenges:

- Performance degradation if overcommitted resources are heavily utilized simultaneously.

Carefully consider your workload patterns and risk tolerance before implementing overcommitment.

Storing Instances' Alternatives

VMs typically store their data on block storage volumes provided by Cinder. However, alternative storage options include:

- **Shared storage:** Multiple VMs can access the same storage pool for data sharing.
- **Object storage (e.g., Swift):** Suitable for storing large amounts of unstructured data like backups and archives.

Understanding Instance Booting

Booting refers to the process of starting a VM. OpenStack offers various boot mechanisms:

- **Cloud-Init:** Provides a standardized way to configure VMs during the boot process.
- **User Data:** Allows users to inject custom scripts or configuration data into the VM during boot.

Planning for Service Recovery

- **Backups:** Regularly back up your VMs and cloud configurations to ensure data recovery in case of failures.
- **High Availability:** Implement strategies like clustering or redundancy to ensure continued service availability even during node failures.
- **Disaster Recovery:** Develop a disaster recovery plan to recover your cloud infrastructure in case of major outages.

By understanding these core concepts and considerations, you can effectively manage your OpenStack compute environment and make informed decisions for optimal resource utilization, security, and service resilience.

OpenStack Compute -

Choice of Hypervisor and Node Segregation

The compute nodes should be separately deployed in the cluster, as it forms the resources part of the OpenStack infrastructure. The compute servers are the heart of the cloud computing service, as they provide the resources that are directly used by the end users; it becomes imperative, then, to give attention to the fact that compute node resources should not be overlooked in terms of processing power, memory, network, and storage capability.

From a deployment…

The compute service components

The compute service is composed of multiple components that take care of receiving the request, and launching and managing the virtual machines. Here is a summary of the various building blocks of the compute service:

- The nova-api service interacts with the user API calls that manage the compute instances. It communicates with the other components of the compute service over the message bus.
- The nova-scheduler is the service that listens to the new instance request on the message bus. The job of this service is to select the best compute node for the new instance.
- The nova-compute service is the process responsible for starting and terminating the virtual machines. This service runs on the compute nodes and listens for new requests over the message bus.

The compute nodes are not provided direct access to the database. This design limits the risk of a compromised compute node providing...

Deciding on the hypervisor

The hypervisor is the heart of your OpenStack compute node. This is called the virtual machine monitor (VMM), which provides a set of manageability functions for virtual machines to access the hardware layer. The amazing part about hypervisors in OpenStack is the wide range of VMMs that it can offer, including KVM, VMware ESXi, QEMU, UML, Xen, Hyper-V, LXC, bare metal, and lately, Docker.

If you already have some experience with one or more of these, it will be better to take a look at how they differ at an architectural level. Currently, the latest OpenStack release at the time of writing this book is Ocata, which has many hypervisor features added or extended. Keep in mind that not all of these support the same features. The Hypervisor Support Matrix (https://wiki.openstack.org/wiki/HypervisorSupportMatrix) is a good reference that can help you to choose what fits your needs.

Obviously...

OpenStack Magnum project

There has been a bit of confusion with container support in OpenStack, especially with implementations such as the Docker driver for Nova and the Magnum project. The OpenStack Magnum project provides Container-as-a-Service capability. So what is the difference between the two approaches? Well, the aim of the Nova Docker driver was to add Docker to be a supported hypervisor for Nova. This enables OpenStack Nova to launch and manage the lifecycle of a Docker container in a fashion similar to other virtual machine instances.

However, it quickly becomes evident that containers, especially the application containers, are not like virtual machines. Hosting an application in containers very often means deploying multiple containers, each running just a single process; these containerized processes then collaborate with each other to provide the complete features of the application. This means that...

Segregating the compute cloud

As your cloud infrastructure grows in size, you need to devise strategies to maintain low latency in the API services and redundancy of your service. To cope with unplanned downtime due to natural forces or based on the hypervisor capability itself, the operator must plan for service continuity.

OpenStack Nova provides several concepts that help you segregate the cloud resources. Each segregation strategy brings in its own advantages and shortcomings. We must have a discussion to understand the tools available with an OpenStack cloud operator to manage the need for scale and availability of compute servers.

Availability zones

The concept of Availability Zones (AZ) in Nova is to group together compute nodes based on fault domains: for...

Overcommitment considerations

In the need for CPU-supporting virtualization we need to understand the number of cores needed, which might affect the CPU power. Remember, for example, that hyper-threading is a highly recommended option for your CPU per compute node in order to double the number of existing cores.

It would be great if you could afford such powerful technology, which is common nowadays. On the other hand, in many cases, the physical compute nodes you purchase might be more powerful than is needed. To avoid such waste, you should keep in mind that sizing your compute nodes is important.

However, this catch-all formula that is applicable in all cases won't be easy to find. You will need to work through three main steps:

1. Estimate a sample calculation for the CPU and RAM size.
2. Use...

Storing instances' alternatives

Compute nodes have been sized with the total CPU and RAM capacity, but we did not cover the disk space capacity. Basically, there are many approaches to doing this, but it might expose other trade-offs: capacity and performance.

External shared file storage

The disks of running instances are hosted externally and do not reside in compute nodes. This will have many advantages, such as the following:

- Ease of instance recovery in the case of compute - node failure
- Shared external storage for other installation purposes

On the other hand, it might present a few drawbacks, such as the following:

- Heavy I/O disk usage affecting the neighboring VM
- Performance degradation due to network latency

Understanding instance booting

Launching an instance on your OpenStack cloud requires interaction with multiple services. When a user requests a new virtual machine, behind the scenes, the user request must be authenticated, a compute node with adequate resources to host the virtual machine must be selected, requests must be made to the image store to get the correct image for the virtual machine, and all the resources required to launch the virtual machine must be allocated. These resources include network connectivity and storage volume allocation.

Understanding the Nova scheduling process

Nova scheduling is one of the critical steps in the process of launching the virtual machine. It involves the process of selecting the best candidate compute node to host a virtual...

Planning for service recovery

One of the most critical tasks for a system administrator or cloud operator is to plan a backup. Building an infrastructure and starting in production without a disaster recovery background is considered highly risky and you will need to start taking immediate action. We may find a bunch of property software in the cloud computing area that does the job, such as the VMware backup solution.

However, backing up open source clouds will not be that easy. OpenStack does not, for instance, support any special tool for backup. As it is merely a collection of components combined to deliver services, an OpenStack operator should think how to map the components used in its infrastructure and prepare a backup strategy for each; the strategy should be easy, efficient, and auto-recovery enabled.

Thus, you should not miss the first question: what do we need to back up and how do we perform such a...

Summary

In this chapter, we have learned about the various scaling and segregation techniques for building a scalable and responsive compute cluster. We understood the instance scheduling, bootup, and initial configuration process.

We discussed containers and container orchestration systems, and the various concepts of the Magnum project. That should give you a clear idea about the variety of hypervisor technologies supported recently by OpenStack.